Q16: Frontier Architectures

Transformer family, MoE, alternatives, and systems

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The question

Q16
What do frontier models use now? Is it still a transformer? Could a better technique replace it?

The qualified answer

- Many frontier LLMs remain decoder-only transformer families.
- The 2017 reference design is not a complete description of production stacks.
- Separate disclosed facts from inference about closed systems.

What remains recognizable

- Token representations + self-attention
- Feed-forward transformations, residual paths, normalization
- Next-token objective for many LLMs

Vaswani et al. (2017)

Within-family changes

- Attention kernels and key/value cache layout
- Positional encoding and long-context methods
- Quantization, batching, compilation, cache policies

A systems optimization is not necessarily an architectural replacement.

MoE changes parameter accounting

- Many experts exist; few are active per token.
- Total parameters $\neq$ activated parameters $\neq$ serving cost.
- Routing/load balance and communication matter.

DeepSeek-AI (2024)

Alternatives and hybrids

- State-space models offer different sequence scaling.
- Mamba uses selective, input-dependent state updates.
- A replacement claim requires matched quality *and* systems evidence.

Gu and Dao (2023)

Comparison protocol

1. Fix tokens, task suite, precision, hardware, and decoding.
2. Report quality, latency, memory, and robustness.
3. Test long contexts and failure cases.
4. State what is unknown for closed systems.

Takeaway

Ask four questions
What component changed? What was held fixed? Which metric improved? What evidence bounds the claim?